

**THE YEAR OF FACEBOOK**

Facebook has certainly come a long way from its modest beginnings in a Harvard dorm – and now that the company has grown public, it's performance has been under severe scrutiny. But it hasn't disappointed, oh no siree.

<http://dgit.in/wvFacebook>

**INTERVIEWING GEORGE R. R. MARTIN**

Game of Thrones, the hit TV show based on the epic fantasy series hardly needs an introduction. Rolling Stone magazine gets some of the most pressing questions answered with an interview with the man behind it all.

<http://dgit.in/wvGRRM>

**CHAT WARS**

It almost seems like an eternity ago – but there was a time when the likes of AIM, ICQ and Yahoo Messenger ruled the Instant Messaging world. MSN Messenger by Microsoft came late to the party and drove AOL mad. Soon, a legal battle ensued. What fun!

<http://dgit.in/wvChatWars>

**HACKING YOUR LIFE**

"Lifehacking" is a term that's relatively recent – and has come to denote any effort in increasing one's personal productivity. While we scramble for any productivity tip that promises to make us more efficient, the concept itself is very, old, going back to the industrial revolution.

<http://dgit.in/wvLifeHacking>

## 4G in India: Challenges and opportunities

While India gears up for the fourth generation of mobile telecommunications, we take a detailed look at the possible challenges and drivers.

– By Kul Bhushan

It has been almost two years since Bharti Airtel launched 4G services in India. Launch of the fourth-generation of mobile telecommunications had come at a time when 3G was still in its nascent stage. While 3G coverage has grown significantly, 4G seems far from becoming mainstream. Though Airtel has rolled out 4G in some more circles, other telecom operators such as Aircel, Videocon Telecom and Reliance Jio are likely to launch their 4G services around year-end or starting next year. Deployment of 4G network in India is certainly not going to be easy as the telecom operators face the challenge to launch a new network along with improving the existing 2G and 3G networks in the country.

### Multiple frequencies

One of the major challenges is the usage of multiple frequencies for rolling out LTE. While co-existence of all radios/spectrum is always possible, it also means that the mobile phone makers need to add support for more radios, which may lead to more costs or complexities. A standard usage of 4G frequency will simplify things for the operators as well as the hardware manufacturers.

Airtel currently offers 4G for mobile on 2300 MHz LTE and it plans to roll out high-speed 4G networks in various circles using FD-LTE (frequency division LTE) technology in the 1,800 MHz band along with its existing TD-LTE (time division LTE) rollout in the 2,300 MHz band, giving it a pan-India 4G footprint.



Even as 2,300 MHz is gaining momentum, the 1800 MHz is considered to be a more "global" spectrum for 4G network. Currently, the 1,800-MHz spectrum is being used to offer only 2G voice services in India but with the government liberalizing spectrum, operators have the option to deploy 4G network on the 1,800 MHz band as well. Telecom operators, however, will need more spectrum from 1,800 MHz to provide smoother LTE service.

Around 108 operators have launched 4G services on 1800 MHz band in 58 nations. 4G on 1800 MHz covers 44 percent of total LTE networks globally. It requires 30 percent lesser towers than on 2,300 MHz.

Similarly, 700MHz and 850MHz bands are considered good for LTE deployment as they offer lower frequency and propagation losses are less. Unlike 1,800 MHz, these bands lack an eco-system and may require a few more years.

### Price and Smartphones

One of the main reasons behind slower adoption of 3G was its steep pricing. 4G faces the same challenge. India is a price-sensitive market. Operators have been wooing customers with low-cost tariff plans for both data and voice. Brief price wars have helped increase 3G adoption in the country. For 4G,

operators will have to launch customized plans for Indian customers.

However, it is going to be really difficult for the telcos as they reel under pressure to maintain their margins as they pay spectrum costs.

"The challenge for telecom service provider in the coming years would be to provide one Rupee per Gigabit over 100 Mbps by 2020 to everyone", says Mr. Vipin Tyagi, Director & Member of Board, Centre for Development of Telematics (C-DOT) while addressing LTE India 2014 conference in Delhi. Users would expect digital action in real time in their daily work and play. Indigenously research development like GPON technology by C-DOT has reduced costs.

"LTE deployment is slowly picking up in the Indian market and we are now neck-to-neck with the global market. While India was almost 10 years late with 2G and almost Six (6) years late with 3G as compared to the